

From temporal leakage to grouped PM_{2.5} prediction

1 AUDIT THE SPLIT

Random overlapping windows

99.90%
context leakage



R^2 : 0.950 random

R^2 : 0.188 complete-date

TRAQID: zero raw-frame overlap

2 CONDITION THE MODEL

Background $B(t, s)$



Visual/context refinement Δ



$$\widehat{\text{PM}}_{2.5} = B + \hat{\Delta}$$

3 TEST GROUPED SHIFTS

54 later Taiwan dates

$$R^2 = 0.706$$

Nested six-site holdout

$$R^2 = 0.702$$

Zero raw-frame overlap